

Write your roll number so legibly that a machine can read it

IIT Bombay
CS 405/6001: GT&AMD
Quiz 1, 2024-25-I
Date: September 6, 2025

Roll No :
e.g., 190040001

Dept.:
e.g., CSE

CS 405/6001: Game Theory and Algorithmic Mechanism Design

Total: 8 + 10 + 12 = 30 marks, Duration: 1 hour, ATTEMPT ALL QUESTIONS

Instructions:

1. This question-and-answersheet booklet contains a total of 5 sheets of paper (10 pages, pages 2 and 10 are blank). Please verify.
2. Write your roll number and department on **every side of every sheet** (except the blank sheet) of this booklet. The first 5 minutes of additional time is given exclusively for this activity. Use only **black/blue ball-point pen**.
3. Write final answers neatly with a pen **only in the given boxes**.
4. Use the rough sheets for scratch works / attempts to solution. **Write only the final solution (which may be a sequence of logical arguments) in a precise and succinct manner in the boxes provided.** Do not provide unnecessarily elaborate steps. The space within the boxes are sufficient for the correct and precise answers.
5. Submit your answerscripts to the teaching staff when you leave the exam hall or the time runs out (whichever is earlier). **Your exam will not be graded if you fail to return the paper.**
6. **This is a closed book, notes, internet exam. No communication device, e.g., cellphones, iPad, smartwatches, etc., is allowed.** Keep them switched off in your bag and keep the bag away from you. If anyone is found in possession of such devices during the exam, that answerscript may be disqualified for evaluation and DADAC may be invoked.
7. One A4 assistance sheet (text **only on one side**) is allowed for the exam.
8. **After you are done with your exam or the exam duration is over, please DO NOT rush to the desk for submitting your paper.** Please remain seated until we collect all the papers, count them, and give a clearance to leave your seat.

Write your roll number so legibly that a machine can read it

IIT Bombay
CS 405/6001: GT&AMD
Quiz 1, 2024-25-I
Date: September 6, 2025

Roll No :
e.g., 190040001

Dept.:
e.g., CSE

Problem 1 (8 points). Answer the following questions.

- (a) Consider the following two-player zero-sum game. Find all the pure strategy Nash equilibria (PSNE) of this game with brief justification. **2 points.**

		Player II	
		L	R
Player I	T	12	8
	B	4	16

- (b) Find all the mixed strategy Nash equilibria (MSNE) of this game. What is the **value** of this game? Recall that a two-player game has a *value* if the minmax and maxmin values of the game are equal. The equal value is then called the value of the game. **2 points.**

Write your roll number so legibly that a machine can read it

IIT Bombay
CS 405/6001: GT&AMD
Quiz 1, 2024-25-I
Date: September 6, 2025

Roll No :
e.g., 190040001

Dept.:
e.g., CSE

- (c) Consider the following game that increases the utility of Player II for every strategy profile by 18. Find all the mixed strategy Nash equilibria (MSNE) of this game. **2 points.**

		Player II	
		L	R
Player I	T	12, 6	8, 10
	B	4, 14	16, 2

- (d) Multiply the utility of the first player in the original game (in Part (a)) by 2, and add 3, to get the following game. Find all the mixed strategy Nash equilibria (MSNE) of this game. What are the equilibria payoffs of the players in this game? **2 points.**

		Player II	
		L	R
Player I	T	27, -12	19, -8
	B	11, -4	35, -16

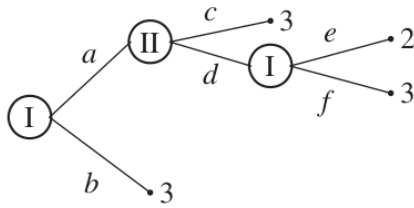
Write your roll number so legibly that a machine can read it

Roll No :
e.g., 190040001

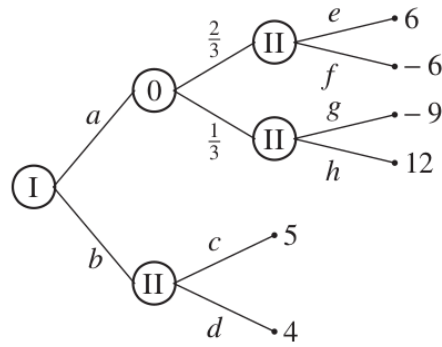
Dept.:
e.g., CSE

IIT Bombay
CS 405/6001: GT&AMD
Quiz 1, 2024-25-I
Date: September 6, 2025

Problem 2 (10 points). Answer the following questions with reference to the two player zero sum games Game A and Game B that appear in the diagram below. Agent 0 is a chance player (e.g., nature) that does not have any utility in this game.



Game A



Game B

(a) Find all subgame perfect Nash equilibria (SPNE) for both the games.

2 × 2 points.

Game A:

Game B:

(b) Represent each of the games in the **normal form**.

2 × 2 points.

Game A:

Write your roll number so legibly that a machine can read it

IIT Bombay
CS 405/6001: GT&AMD
Quiz 1, 2024-25-I
Date: September 6, 2025

Roll No :
e.g., 190040001

Dept.:
e.g., CSE

Game B:

- (c) Are there other pure strategy Nash equilibria (PSNE) except the ones above? If yes, list them for each game. Write “none” if no other PSNE exists. **2 × 1 points.**

Game A:

Game B:

Write your roll number so legibly that a machine can read it

IIT Bombay
CS 405/6001: GT&AMD
Quiz 1, 2024-25-I
Date: September 6, 2025

Roll No :
e.g., 190040001

Dept.:
e.g., CSE

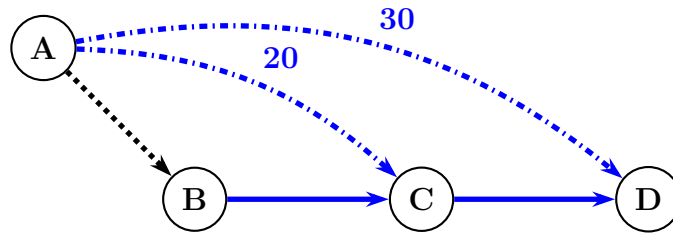
Problem 3 (12 points). City of K is expanding its metro railway infrastructure. Initially, the existing line is the blue line connecting hubs B, C, and D as shown in the figure below. There are also two bus routes AC and AD as shown, each of which charges a fixed ticket price of 20 and 30 respectively. At this point, the yellow line, shown as dashed in AB, is not ready for use.

There are two major residential areas: one near A with a population size of 10 and another near B of size 4. All the workplaces are located at C and D. The passengers from B has only the blue line as their option for traveling irrespective of the prices. Their daily travel plan is as follows: 2 get down at C and the remaining 2 travel to D for work.

The travel destinations for the passengers from A are: C for 2 and D for the remaining 8. We assume that in the context of travel in this problem, every passenger travels only in the direction of the arrows of this figure. Therefore, the 2 people destined for C has only one choice before the yellow line opens: travel via AC. However, the remaining 8 has a choice of using either ACD or AD.

Every passenger is rational and intelligent. They aim to minimize the travel **cost** which is given by $kx + p$ for a travel segment, where k is a constant for a type of travel ($k = 1$ for metro and $k = 2$ for bus), x is the number of passengers traveling on that path, and $p \geq 0$ is the ticket price of that segment of travel. Intuitively, kx is the **human cost** of this travel and p is the **monetary cost**. The costs are additive over segments, i.e., the costs of a passenger traveling in two segments add up.

Being the oldest metro line, the blue line has a safety limit of carrying 7 passengers, both in BC and CD. Hence, in addition to the 4 and 2 passengers of area B, it can safely carry 3 and 5 more passengers in BC and CD respectively. It can carry more, but should be avoided if possible. This problem asks you how to set the prices of the metro tickets in the BC and CD segments so that this safety limit is maintained in the equilibrium before and after the opening of the yellow line. Assume that the newly built yellow line AB does not have any such maximum occupancy constraint. Similarly, there are sufficient number of buses to carry passengers, and the maximum occupancy constraint does not apply there either.



Provide only the necessary answers to the following questions. DO NOT show the calculations.

- (a) Before the yellow line segment AB was opened, what would be the range of ticket prices in the CD segment such that the safety limit is maintained? Provide a range of prices explaining which price will lead to what passenger flow. Only provide the range of prices that changes the passenger flow.

2 points.

Write your roll number so legibly that a machine can read it

IIT Bombay
CS 405/6001: GT&AMD
Quiz 1, 2024-25-I
Date: September 6, 2025

Roll No :
 e.g., 190040001

Dept.:
 e.g., CSE

- (b) Before the yellow line segment AB was opened, what price of CD segment will make the passenger load equal in the two bus routes maintaining the safety limits? What will be that equal passenger load? **2 points.**

- (c) After the yellow line segment AB is opened, assume that the ticket prices are zero for the segment AB as a promotional offer. What prices of BC and CD segments will maintain the safety limits? Show the region of feasible prices in a diagram with p_{BC} as x-axis and p_{CD} as y-axis. Mark the endpoints of the regions clearly (it is okay to leave the numbers in fractions, but the points should be clearly marked). Only provide the range of prices that changes the passenger flow. **4 points.**

Write your roll number so legibly that a machine can read it

IIT Bombay
CS 405/6001: GT&AMD
Quiz 1, 2024-25-I
Date: September 6, 2025

Roll No :
e.g., 190040001

Dept.:
e.g., CSE

-
- (d) After the yellow line segment AB is opened, what prices of BC and CD segments will maintain the safety limits and make the passenger load in AB, AC, and AD equal? **2 points.**

- (e) After the yellow line segment AB is opened, what minimum equal prices of BC and CD segments will maintain the safety limits, i.e., $p_{BC} = p_{CD}$ and also the minimum such value? **2 points.**

END OF QUESTION PAPER. GOOD LUCK!

